



## 10. Threat Intelligence

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## Top stories

Canvas access restored after cyberattack by ShinyHunters >



 The Verge

Canvas is online again after ShinyHunters threaten to leak schools' data

1 day ago

 Cyber Daily

Aussie schools breach: The Instructure hack 'transcends an isolated IT incident'



1 day ago

 BleepingComputer

Canvas login portals hacked in mass ShinyHunters extortion campaign

1 day ago



 Bitdefender

Technical Advisory: ShinyHunters Breach of Instructure Canvas LMS

3 days ago



 WBIW

ShinyHunters cyberattack sparks nationwide alarm after Canvas breach

6 hours ago



Final exam week in the US for the Spring 2026 semester typically occurs in early to **mid-May**

**ShinyHunters** is a **black-hat** criminal **hacker** and **extortion** group that has been active since 2020 or earlier,<sup>[2][3]</sup> and is said to have been involved in a significant number of **data breaches**.<sup>[2]</sup>

# Position of Cyber Threat Intelligence

|                                 | Focus                                                             | Goal                                                                                | Example                                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Threat Modeling                 | System architecture, attack surfaces, and possible attack paths   | Identify how attackers <i>could</i> compromise a system before or during deployment | “The Canvas learning system may be vulnerable to credential stuffing and API abuse because it exposes internet-facing authentication services.”                                                 |
| Cyber Threat Intelligence (CTI) | Real-world threat actors, campaigns, TTPs, and active threats     | Understand what attackers are <i>actually doing</i> and how threats are evolving    | “Threat intelligence reports indicate that the ShinyHunters group claims to have breached systems related to the Canvas learning ecosystem and leaked user data online.”                        |
| Risk Assessment                 | Likelihood and business impact of threats to the organization     | Determine which risks are most significant and should be prioritized                | “Because Canvas stores student records and authentication data, a potential compromise could lead to operational interruption, high reputational damage and privacy impact for the university.” |
| Risk Management                 | Organizational decisions and controls for handling risk over time | Select and implement strategies to reduce, transfer, accept, or monitor risk        | “The university decides to enforce MFA, improve monitoring, conduct incident response preparation, and allocate additional funding to reduce Canvas-related cyber risk.”                        |

# Data vs Information vs Intelligence

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## **Data** (Raw Facts)

- ◇ Unprocessed, isolated observations:
  - ◇ Temperature: 18°C
  - ◇ Humidity: 90%
  - ◇ Rain probability: 80%
  - ◇ Dark clouds observed



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  - ◇ “There is a high chance of rain this afternoon.”

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- ◇ Information is analyzed with context, goals, and consequences:
  - ◇ “I’ll be outside this afternoon  
→ I should bring an umbrella.”

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**Data** tells you what happened, **information** tells you what is happening, **intelligence** tells you what to do next.

# Data vs Information vs Intelligence

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| Stage               | Description                | Cybersecurity Example                                          |
|---------------------|----------------------------|----------------------------------------------------------------|
| <b>Data</b>         | Raw signals                | Logs, IP addresses, packet captures                            |
| <b>Information</b>  | Processed & contextualized | “IP 1.2.3.4 attempted 500 logins in 5 minutes”                 |
| <b>Intelligence</b> | Actionable decision        | “This is likely a brute-force attack → block IP and alert SOC” |



# Types of Threat Intelligence



**Different levels/types of intelligence serve different audiences and decisions.**

## 1. STRATEGIC INTELLIGENCE

*Why it matters*



High-level, long-term perspective on the threat environment and its impact on the organization.



### FOCUS

Trends, risks, and business impact that drive decisions and investments.



### USED BY

Executives, CISO, Board, Senior Leadership



### EXAMPLE

Ransomware attacks against the education sector increased 40% this year, causing significant financial and reputational risk.

Purpose: Guide strategy and resource allocation

## 2. OPERATIONAL INTELLIGENCE

*How campaigns work*



Intelligence about threat actors, their campaigns, targets, and objectives.



### FOCUS

Who is attacking, what campaigns they run, and what they aim to achieve.



### USED BY

Security Managers, Threat Intel Analysts, Incident Response Leads



### EXAMPLE

APT29 is targeting government and defense organizations with phishing and credential theft campaigns since Jan 2024.

Purpose: Understand the threat landscape and prepare defenses

### 3. TACTICAL INTELLIGENCE

*How attacks are performed*



Knowledge of adversary techniques, tactics, and procedures (TTPs) used in attacks.



#### FOCUS

How adversaries operate within systems and networks.



#### USED BY

SOC Analysts, Threat Hunters, Detection Engineers



#### EXAMPLE

Attackers use brute-force logins, lateral movement via SMB, and privilege escalation using PowerShell.

Purpose: Improve detection, hunting and response effectiveness

### 4. TECHNICAL INTELLIGENCE

*What to detect or block*



Specific, actionable indicators of compromise (IOCs) about malicious artifacts and infrastructure.



#### FOCUS

Exact artifacts that can be detected or blocked by security tools.



#### USED BY

SIEM, IDS/IPS, Firewall, EDR, Security Tools (Automation)



#### EXAMPLE

- Malicious IP: 203.0.113.45
- Domain: update-secure[.]com
- File Hash (SHA256): 3f2a6c8e...e9b7d2a1
- Malware: agent.exe

Purpose: Enable immediate detection and automated blocking

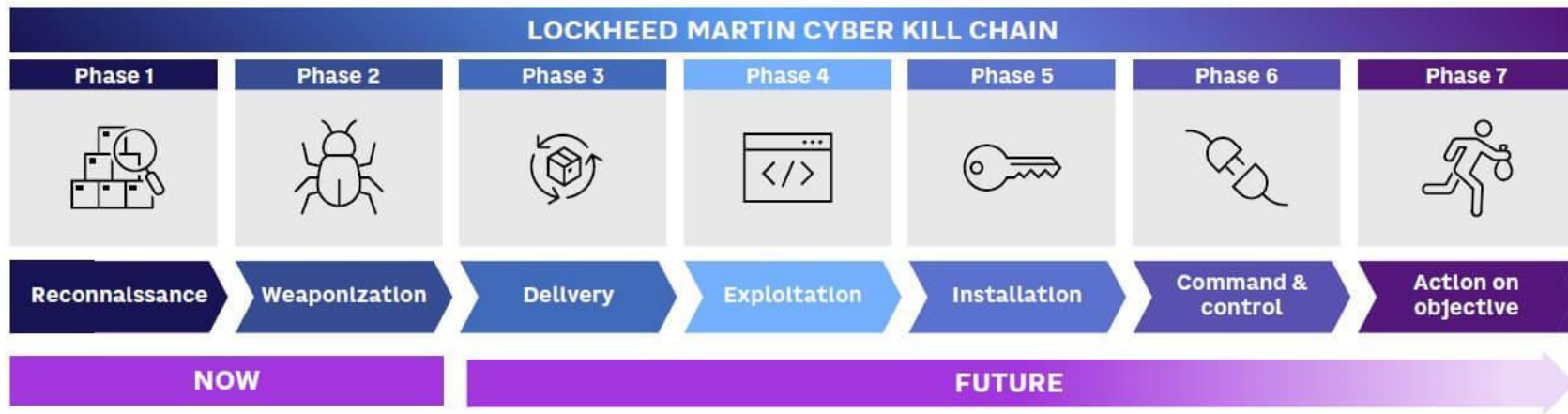


# CTI Frameworks

## Purpose of CTI Frameworks

- ◇ CTI frameworks help answer:
  - ◇ **How attacks happen** → structure attacker behavior
  - ◇ **Where to detect/defend** → identify security controls
  - ◇ **How to communicate threats** → standardize language
- ◇ Three Key Frameworks
  - ◇ **Lockheed Martin Cyber Kill Chain** → *Attack stages*
  - ◇ **MITRE ATT&CK** → *Adversary behaviors (TTPs)*
  - ◇ **Diamond Model** → *Adversary relationships*

# CTI Frameworks



Source: Arthur D. Little

## ◆ Pros

- ◆ Simple and intuitive
- ◆ Great for **high-level understanding**
- ◆ Helps identify **where to stop attacks early**

## ◆ Cons

- ◆ Too **linear** (real attacks are not)
- ◆ Limited detail on attacker techniques
- ◆ Less useful for modern, complex threats



# CTI Frameworks

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## MITRE ATT&CK

- ◇ A **knowledge base of adversary behaviors (TTPs)** observed in real attacks.
  - ◇ **Tactics** → attacker goals (e.g., Initial Access)
  - ◇ **Techniques** → how they achieve it
  - ◇ **Procedures** → specific implementations

# CTI Frameworks

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- ◇ A **knowledge base of adversary behaviors (TTPs)** observed in real attacks.
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  - ◇ **Techniques** → how they achieve it
  - ◇ **Procedures** → specific implementations
- ◇ **Pros**
  - ◇ Very **detailed and practical**
  - ◇ Widely adopted in industry
  - ◇ Supports:
    - ◇ Threat hunting
    - ◇ Detection engineering
    - ◇ Red/blue teaming
- ◇ **Cons**
  - ◇ Can be **overwhelming for beginners**
  - ◇ Not a full lifecycle model
  - ◇ Requires effort to map data to techniques

# CTI Frameworks

Diamond Model models an intrusion as a **relationship between four elements**:

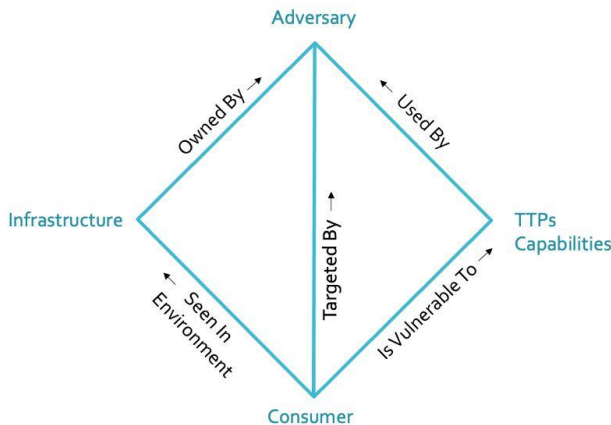
- ◇ **Adversary** (who)
- ◇ **Capability** (tools/malware)
- ◇ **Infrastructure** (IP/domain)
- ◇ **Victim** (target)

## ◇ Pros

- ◇ Great for **intelligence analysis & investigation**
- ◇ Encourages **linking evidence** across incidents
- ◇ Supports attribution and campaign tracking

## ◇ Cons

- ◇ Less direct for detection engineering
- ◇ More **analytical than operational**
- ◇ Requires experience to apply effectively



# CTI Frameworks

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- ◆ These frameworks are **complementary, not competing**.

- ◆ Kill Chain → *Where are we in the attack?*

- ◆ ATT&CK → *How is the attacker operating?*

- ◆ Diamond Model → *Who and what is involved?*

- ◆ **Practical Workflow**

- ◆ Use **Kill Chain** → understand attack progression

- ◆ Use **ATT&CK** → map techniques & detect

- ◆ Use **Diamond Model** → analyze relationships



# Threat Actors and Motivations

## ◆ **Nation-State Actors**

- ◆ Also known as **Advanced Persistent Threats (APTs)**
- ◆ Backed by governments (state-sponsored)
- ◆ Highly skilled and well-funded

## ◆ **Motivation**

- ◆ **Espionage (primary)**
- ◆ Intellectual property theft
- ◆ National security intelligence



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## ◆ **Characteristics**

- ◆ Long-term access (**persistence**)
- ◆ Stealthy (avoid detection)
- ◆ Highly sophisticated techniques

## ◆ **Example Behaviors**

- ◆ Silent data exfiltration
- ◆ Zero-day exploitation
- ◆ Supply chain compromise



Get in, stay hidden, extract value, and leave without being noticed.

# Threat Actors and Motivations

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- ◆ **Cybercriminals** (Financially Motivated)
  - ◆ Most common threat actors
  - ◆ Can be highly organized and well-funded
- ◆ **Motivation**
  - ◆ Financial gain (primary)



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- ◆ **Motivation**

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- ◆ **Characteristics**

- ◆ Opportunistic or targeted
- ◆ Scalable attacks (e.g., phishing)
- ◆ May reinvest profits into better tools

- ◆ **Example Attacks**

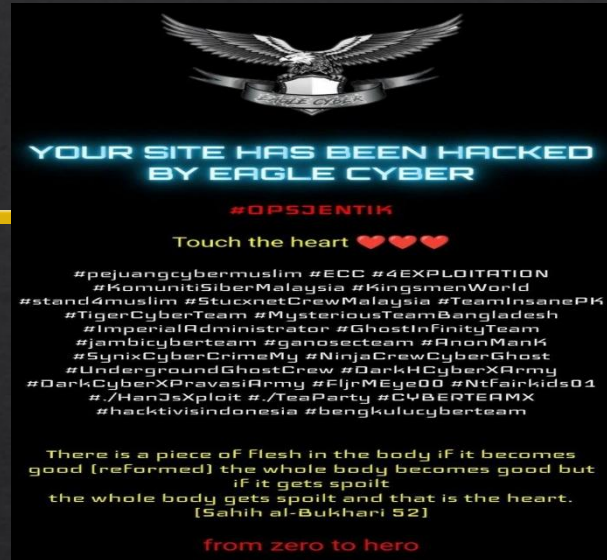
- ◆ Ransomware
- ◆ Phishing campaigns
- ◆ Business Email Compromise (BEC)
- ◆ Supply chain attacks

Maximize profit with minimum effort and acceptable risk.

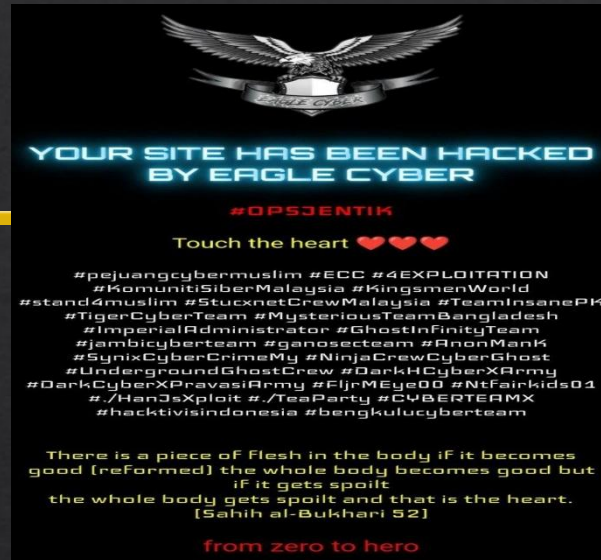


# Threat Actors and Motivations

- ◆ **Hacktivists** (Ideology-Driven Attackers)
  - ◆ Ideology-driven attackers
  - ◆ Not typically state-sponsored
  - ◆ Aim to **influence public opinion or policy**
- ◆ Motivation
  - ◆ **Political or social causes**
  - ◆ Activism (digital protest)



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- ◆ Activism (digital protest)

## ◆ Characteristics

- ◆ Focus on **visibility and impact**
- ◆ Less concerned with stealth
- ◆ Attacks often **public and symbolic**

## ◆ Common Attacks

- ◆ DDoS attacks (service disruption)
- ◆ Website defacement
- ◆ Data leaks (to expose information)

Make noise, attract attention, push a message.

# Threat Actors and Motivations

- ◆ **Script Kiddies** (Low-Skill Attackers)

- ◆ Inexperienced attackers
- ◆ Use **pre-built tools/scripts**

- ◆ **Motivation**

- ◆ Curiosity / learning
- ◆ Reputation or bragging rights
- ◆ Occasionally small financial gain





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## ◇ **Characteristics**

- ◇ Limited technical skills
- ◇ Depend on publicly available tools
- ◇ Opportunistic rather than targeted

## ◇ **Common Attacks**

- ◇ Basic website exploitation
- ◇ Vulnerability scanning
- ◇ Simple DoS attacks

Use what's available, learn by doing, sometimes cause unexpected damage.



# CTI Users

- ◆ The Security Operational Center (**SOC**) Problem
- ◆ **Key Challenge**: Too many alerts, not enough people

- ◆ **Reality**

- ◆ SOC receives **1000+ alerts per day**
- ◆ Analysts have **only minutes per alert**
- ◆ High risk of:
  - ◆ Alert fatigue
  - ◆ Missed attacks
  - ◆ Burnout

- ◆ **Without CTI**

- ◆ Treat all alerts equally
- ◆ Slow investigation
- ◆ Inefficient response

- ◆ **Key Question**

- ◆ “**Which alerts actually matter?**”



# CTI Users

- ◆ How CTI Helps the SOC
- ◆ **Prioritize and automate using CTI**
- ◆ **Two Key Functions**
  - ◆ **1. Prioritization**
    - ◆ Use intelligence to identify **high-risk alerts**
    - ◆ Focus analyst time where it matters
  - ◆ **2. Automation**
    - ◆ Auto-block known threats
    - ◆ Auto-enrich alerts with context



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- ◇ **Example**
  - ◇ Alert: suspicious file download
  - ◇ CTI says: file seen in past attacks  
**Escalate immediately**
- ◇ **Outcome**
  - ◇ Faster triage
  - ◇ Reduced workload
  - ◇ Better detection accuracy

# CTI Users

- ◆ What Intelligence the SOC Uses:
  - ◆ **Technical Intelligence**
    - ◆ **Indicators of Compromise (IOCs)**
      - ◆ Malicious IPs
      - ◆ File names / hashes
      - ◆ Used for **quick decisions**
  - ◆ **Tactical Intelligence (TTPs)**
    - ◆ Focus on **procedures**:
      - ◆ What steps attackers follow
      - ◆ How attacks unfold
- SOC needs to recognize attack patterns—not just single indicators



- ◆ **Analogy**
  - ◆ **Tactic** → Celebrate a birthday
  - ◆ **Technique** → Have a cake
  - ◆ **Procedure** → Recipe to bake the cake



# CTI Users



- ◇ When **Incident Response Team** uses CTI

- ◇ **Two** Key Moments for CTI Use

- ◇ **During an Attack (Detection & Triage)**

- ◇ “Is this a real attack or just noise?”

- ◇ Validate alerts

- ◇ Prioritize investigation

- ◇ Understand likely attack behavior

- ◇ **After an Attack (Impact & Recovery)**

- ◇ “What damage has been done?”

- ◇ Assess impact

- ◇ Identify attacker

- ◇ Guide recovery actions

IR teams need **different intelligence at different stages**

# CTI Users



- ◇ Post-Attack: **Dark Web & Forum Intelligence**
- ◇ What has been exposed?
- ◇ **Dark Web / Forum Intelligence**
- ◇ Monitor:
  - ◇ Dark web marketplaces
  - ◇ Hacker forums
  - ◇ Ransomware leak sites
- ◇ **What IR Teams Look For**
  - ◇ Company data being sold
  - ◇ Credentials or databases leaked
  - ◇ Mentions of the organization

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- ◆ **Why It Matters**

- ◆ Determine **risk level**:

- ◆ Financial (fines, ransom)
- ◆ Reputational damage

- ◆ Help identify:

- ◆ Which threat actor is responsible

- ◆ Most organizations **outsource dark web monitoring** due to complexity

# CTI Users

◇ IR Use CTI to guide **targeted investigation**: What has been exposed?

◇ **Inputs from CTI**

◇ ◆ **Technical (IOCs)**

- ◇ File hashes
- ◇ Malicious IPs
- ◇ Domains

◇ ◆ **Tactical (TTPs)**

- ◇ Attack steps
- ◇ Malware behavior
- ◇ Lateral movement patterns

**Example:** Malware detected on one endpoint

◇ CTI tells you:

- ◇ Type of malware
- ◇ Expected behavior

◇ IR team can:

- ◇ Identify affected systems
- ◇ Search for related indicators
- ◇ Contain spread faster



# CTI Users

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- ◇ Speed, Prioritization & Intelligence Balance
- ◇ IR teams must act **fast under pressure**
- ◇ **How CTI Helps**
  - ◇ **Prioritization**
    - ◇ Focus on **most relevant alerts**
    - ◇ Based on:
      - ◇ Known attackers
      - ◇ Likely attack methods
  - ◇ **Speed**
    - ◇ Provide **actionable intelligence only**
    - ◇ Avoid overwhelming data feeds

# CTI Users

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## All Level Intelligence Needed

| Level       | Why IR Needs It             |
|-------------|-----------------------------|
| Strategic   | Understand threat landscape |
| Operational | Know active campaigns       |
| Tactical    | Understand attack methods   |
| Technical   | Investigate and respond     |

# CTI Users

- ◇ Why Leadership Uses CTI?
- ◇ Keep the organization **within risk tolerance** with **efficient spending**



- |                                                                                                                                                                                                                                                             |                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>◇ <b>What Leadership Needs from CTI</b><ul style="list-style-type: none"><li>◇ Understand <b>relevant threats</b></li><li>◇ Assess <b>business impact</b></li><li>◇ Decide <b>where to invest</b></li></ul></li></ul> | <ul style="list-style-type: none"><li>◇ <b>What They Don't Need</b><ul style="list-style-type: none"><li>◇ Raw logs</li><li>◇ File hashes</li><li>◇ Low-level technical details</li></ul></li></ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Leadership needs to know: What is happening, why it matters, and what we should do

# CTI Users

- ◆ Non-Technical Leadership (Business Focus)
- ◆ CEO, CFO, senior leadership (non-security)



- ◆ **What They Care About**

- ◆ Business impact
- ◆ Financial risk
- ◆ Reputation
- ◆ Industry trends

- ◆ **Example (Effective):** “In the last 2 weeks, ransomware groups attacked 5 similar companies, costing ~\$X each.”
- ◆ Leads to:
  - ◆ Budget approval
  - ◆ Policy changes

CTI makes **risk tangible and measurable**



# CTI Users

- ◆ Technical Leadership (CISO / Security Leaders)

- ◆ CISO, security leadership

- ◆ **What They Care About**

- ◆ Relevance to organization
- ◆ Threat actors targeting them
- ◆ Attack methods (high-level)
- ◆ Frequency of attacks
- ◆ Financial impact

- ◆ **Example Questions**

- ◆ Are we a target? Why?
- ◆ What attacks are trending?
- ◆ What controls should we implement?

- ◆ **How CTI Helps**

- ◆ Map threats to:
  - ◆ Industry
  - ◆ Geography
  - ◆ Technology stack

Tell me what matters to *our organization*, not just what exists

# CTI Users

- ◆ Decision-Making & Resource Allocation

- ◆ **Investment**

- ◆ Where to spend security budget
- ◆ Which controls to prioritize

- ◆ **Workforce Strategy**

- ◆ Hire more staff
- ◆ Train existing team
- ◆ Automate security tasks



- ◆ **Requirements for Good CTI**

- ◆ Relevant (to the organization)
- ◆ Trusted (credible sources)
- ◆ Concise (executive-friendly)
- ◆ Actionable (includes solutions)

# CTI Users

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| User       | CTI Purpose                         |
|------------|-------------------------------------|
| SOC        | Prioritize alerts                   |
| IR         | Investigate & respond               |
| Leadership | Make decisions & allocate resources |

# CTI Life-Cycle





# Step1: Planning & Direction

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- ◇ What is **Planning & Direction**?
- ◇ The stage where you define **what you need to protect and why**
  
- ◇ **Key Objectives**
  - ◇ Set **CTI goals**
  - ◇ Identify **critical assets (crown jewels)**
  - ◇ Define **intelligence requirements**
- ◇ **Common Mistake**
  - ◇ Jumping straight to **data collection or tools** without planning
  - ◇ Good CTI starts with **understanding priorities**, not data

# Step1: Planning & Direction

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- ◆ Identify “**Crown Jewels**”
- ◆ The **most valuable assets** in your organization

- ◆ **Common Examples**

- ◆ Customer data
- ◆ Financial data
- ◆ Intellectual property (IP)
- ◆ Critical systems

- ◆ **Key Questions**

- ◆ What happens if this is stolen or lost?
- ◆ What is the **business impact**?

- ◆ **Example:** If protecting IP

- ◆ Who would target it?
- ◆ How do they attack?
- ◆ How often does this happen?

You can't protect everything equally—  
prioritize what matters most

# Step1: Planning & Direction

- ◇ Define Intelligence Requirements
- ◇ Different teams need **different intelligence**
- ◇ **Identify Stakeholders**
  - ◇ SOC → technical & fast intelligence
  - ◇ Incident Response → detailed & investigative
  - ◇ Leadership → strategic & business-focused
- ◇ **Define Requirements**
  - ◇ What intelligence is needed?
  - ◇ Who needs it?
  - ◇ How often?
  - ◇ In what format?

## ◇ Example

- ◇ Researchers (IP owners):
  - ◇ Need **context & awareness**
  - ◇ Don't need raw indicators

**CTI must be tailored to the user, not one-size-fits-all**



# Step1: Planning & Direction

## ◇ Consider the Context

- ◇ Has the organization:
  - ◇ Recently been attacked?
  - ◇ Entered new regulations?
  - ◇ Changed business strategy?

## ◇ Understand Existing Environment

- ◇ Current defenses
- ◇ Security tools
- ◇ Team structure

## ◇ Why This Matters

- ◇ Avoid duplication
- ◇ Align CTI with existing capabilities
- ◇ Improve collaboration

## ◇ **Final Warning:** Skipping this step leads to:

- ◇ Irrelevant intelligence
- ◇ Wasted resources
- ◇ Poor decision-making

Planning & Direction =

**What to protect + Who needs intelligence + Why it matters**

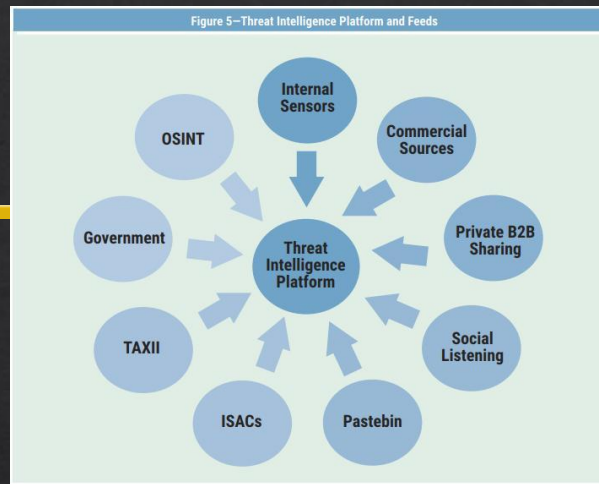


# Step2: Collection

- ◆ What is **Collection**?
- ◆ Gathering **raw data and information** to later turn into intelligence

- ◆ **Internal Sources (Most Valuable)**
  - ◆ System logs
  - ◆ Endpoint Detection & Response (EDR)
  - ◆ Past incidents
- ◆ Directly relevant to your organization

- ◆ **External Collaboration**
  - ◆ Peer organizations
  - ◆ Government agencies like
    - ◆ CISA
    - ◆ NCSC
- ◆ Share threat insights across industries



# Step2: Collection

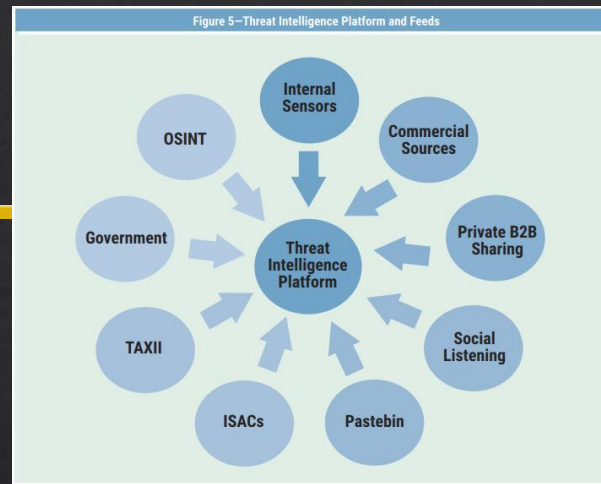
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- ◆ **Public Sources (OSINT)**

- ◆ Industry blogs (attack news, vulnerabilities)
- ◆ Social media (fraud discussions, threats)
- ◆ Developer forums (new vulnerabilities)

- ◆ **Threat Feeds**

- ◆ Free or commercial feeds
- ◆ Provide:
  - ◆ IOCs (IPs, domains, hashes)
  - ◆ Known malicious activity



# Step2: Collection

## ◆ **Key Challenge**

- ◆ Too many sources → too much data
- ◆ What You Need to Consider

## ◆ **Automation**

- ◆ Automate data collection where possible
- ◆ Use tools to ingest:
  - ◆ Threat feeds
  - ◆ Logs
  - ◆ OSINT

## ◆ **Scale**

- ◆ Not all data is useful
- ◆ Focus on:
  - ◆ Relevant sources
  - ◆ High-priority threats

## ◆ **Trust & Quality**

- ◆ Not all sources are reliable
- ◆ Ask:
  - ◆ Is this source credible?
  - ◆ Is the data accurate?

**Getting the right data from the right sources (not all data)**

# Step2: Collection

- ◆ What is the **Dark Web** in CTI?
- ◆ A central gathering place for cybercriminals where intelligence can be collected



- |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>◆ <b>What You Can Find</b></li><li>◆ <b>Data for Sale (Most Common)</b><ul style="list-style-type: none"><li>◆ Breach credentials</li><li>◆ Personal or corporate data</li></ul></li><li>◆ Often how organizations discover a breach</li></ul> | <ul style="list-style-type: none"><li>◆ <b>Threat Actor Discussions</b><ul style="list-style-type: none"><li>◆ New vulnerabilities</li><li>◆ Exploit techniques</li><li>◆ Shared attack tools</li><li>◆</li></ul></li></ul> | <ul style="list-style-type: none"><li>◆ <b>Rare but Valuable</b><ul style="list-style-type: none"><li>◆ Mentions of your organization</li><li>◆ Planned or potential attacks</li></ul></li><li>◆ <b>Key Use Cases</b><ul style="list-style-type: none"><li>◆ Post-breach monitoring</li><li>◆ Early signals of emerging threats</li><li>◆ Understanding attacker techniques</li></ul></li></ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



# Step2: Collection

## ◆ Limitations of Dark Web Intelligence

### ◆ **Not Predictive** (Most Cases)

- ◆ Data appears after a breach
- ◆ May be **old** or recycled

### ◆ Biased Toward Cybercrime

- ◆ Mostly **financially motivated actors**
- ◆ Nation-state (espionage) actors rarely appear

### ◆ Data Quality Issues

- ◆ Outdated information
- ◆ False or misleading posts

## ◆ Access Challenges

### ◆ Some forums:

- ◆ Easy to access → can automate scraping

### ◆ Others:

- ◆ Require trust & participation
- ◆ Hard to maintain access

Often **outsourced to specialist providers**



# Step3: Processing

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- ◇ Converting **raw collected data** into a **usable format for analysis**

- ◇ **Why It's Needed**

- ◇ Data is:

- ◇ Large in volume
- ◇ Unstructured
- ◇ Inconsistent

- ◇ Reduce and organize data into something workable

- ◇ **Examples**

- ◇ Translate foreign-language content
- ◇ Standardize formats (logs, feeds)
- ◇ Structure unorganized data

# Step3: Processing

## Typical Processing Activities

### ◆ Translation

- ◆ Dark web / global sources → different languages
- ◆ Convert into understandable format

### ◆ Deduplication

- ◆ Remove repeated data
- ◆ Reduce noise and redundancy

### ◆ Normalization

- ◆ Convert data into consistent formats
- ◆ Example:
  - ◆ IP formats
  - ◆ Log structures

### ◆ Filtering

- ◆ Remove irrelevant data
- ◆ Keep only what matches:
  - ◆ Your organization
  - ◆ Your intelligence goals

**Processing improves clarity, efficiency, and usability**

# Step3: Processing

## Evaluate Data Quality

- ◇ Ask:
  - ◇ Is the source **reliable**?
  - ◇ Can we **trust** the data?
  - ◇ Do we have **multiple sources**?
- ◇ Check Relevance
  - ◇ Is it relevant to:
    - ◇ Your organization?
    - ◇ Your investigation?

- ◇ **Example:** Found leaked credentials with same name as executive. Need to verify:

- ◇ Same person or coincidence?
- ◇ Password reuse risk?

## ◇ Automation

- ◇ Processing can be:
  - ◇ Manual (analysts)
  - ◇ Automated (tools, pipelines)
- ◇ More automation =
  - ◇ Faster processing
  - ◇ Ability to handle more data



# Step4: Analysis



- ◇ What is **Analysis**?
- ◇ The process of converting **processed data** → **actionable intelligence**

- ◇ **Why It Matters**

- ◇ Data alone is not useful
- ◇ Analysis provides:
  - ◇ Meaning
  - ◇ Context
  - ◇ Decisions

- ◇ **Who Does It?**

- ◇ Human analysts (critical thinking required)
- ◇ Tools can assist, but **humans interpret**

**Analysis is where intelligence is created—not collected**

# Step4: Analysis



## ◆ **Key Questions Analysts Ask**

### ◆ **About the Data**

- ◆ What information do I have?
- ◆ Is it complete?
- ◆ How much data is available?

### ◆ **About the Source**

- ◆ Who provided this information?
- ◆ Is it trusted?
- ◆ Victim or attacker?

### ◆ **About Timing**

- ◆ Is this recent or old?

### ◆ **About Context**

- ◆ What type of attack is this?
- ◆ What is the broader situation?

### ◆ **About Gaps**

- ◆ What don't I know?
- ◆ What is missing?
- ◆ What else do I need?

**Good analysis questions assumptions, not just data**

# Step4: Analysis

## ◆ What Analysis Produces

- ◆ Explanation → What is happening?
- ◆ Attribution → Who is responsible?
- ◆ Prediction → What will happen next?
- ◆ Recommendation → What should we do?

## ◆ Example Outputs

- ◆ “Likely ransomware campaign targeting our sector”
- ◆ “Low confidence claim—possible bluff”
- ◆ “Prioritize these systems for investigation”





# Step4: Analysis



## ◆ What Analysis Produces

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## ◆ Example Outputs

- ◆ “Likely ransomware campaign targeting our sector”
- ◆ “Low confidence claim—possible bluff”
- ◆ “Prioritize these systems for investigation”

| User       | Benefit                     |
|------------|-----------------------------|
| SOC        | Better alert prioritization |
| IR         | Faster investigation        |
| Leadership | Informed decisions          |

Intelligence is only valuable if it leads to action



# Step5: Dissemination

- ◆ What is **Dissemination**?
- ◆ Getting the right intelligence to the right people, in the right way

- ◆ Key Questions

- ◆ Who needs the intelligence?
  - ◆ What do they need?
  - ◆ How often do they need it?
  - ◆ In what format do they consume it?
- ◆ Same intelligence  $\neq$  same presentation



| User       | Format                            |
|------------|-----------------------------------|
| SOC        | Real-time alerts, dashboards      |
| IR         | Detailed reports, IOCs + TTPs     |
| Leadership | Summaries, risk-focused briefings |

# Step5: Dissemination

## ◆ What Good Dissemination Looks Like

- ◆ Relevant → tailored to the user
- ◆ Timely → delivered when needed
- ◆ Clear → easy to understand
- ◆ Actionable → supports decisions

## ◆ Common Problems

- ◆ Too much technical detail
- ◆ Too much data (overload)
- ◆ Not actionable

## ◆ Key Test

- ◆ “Can the user **take action immediately** based on this?”

## ◆ Example

- ◆ SOC → “Block this IP now”
- ◆ Leadership → “This threat could cost \$X → invest in control Y”



# Step6: Feedback

- ◆ What is **Feedback**?
- ◆ Collecting input from users to **improve the CTI process**

## **What to Ask**

### ◆ **Dissemination**

- ◆ Is the information useful?
- ◆ Is it delivered at the right time?
- ◆ Is the format appropriate?

### ◆ **Trust & Quality**

- ◆ Is the intelligence trusted?
- ◆ Is there duplicate or inaccurate data?  
If not → revisit **processing**

### ◆ **Coverage**

- ◆ Is intelligence comprehensive enough?  
If not → revisit **collection**

### ◆ **Continuous Improvement Loop**

- ◆ Feedback → adjust process
- ◆ Needs change over time
- ◆ CTI must evolve with the business

# Conclusion

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- ◇ **Data → Information → Intelligence**
  - ◇ CTI transforms raw data into actionable decisions
- ◇ **Three Frameworks**
  - ◇ MITRE ATT&CK → attacker behavior
  - ◇ Cyber Kill Chain → attack stages
  - ◇ Diamond Model → relationships
- ◇ Different Users, Different Needs
- ◇ CTI Lifecycle (Continuous Process)

**Acknowledgement: Learning Threat Intelligence (Linkedin Course) by Karla Reffold**



# Additional readings

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- ◆ Pyramid of Pain

- ◆ <https://www.attackiq.com/glossary/pyramid-of-pain/>

- ◆ CTI list

- ◆ <https://github.com/hslatman/awesome-threat-intelligence>